PromptLab: Optimizing LLM Slop

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Turning Prompt Engineering into an Optimization Problem

Slop = high fluency
but low informational value

Characteristics:

- Redundant phrasing
- Generic abstractions
- Low specificity
- Predictable structure

➔ Treat prompt design as an optimization problem

- Prompt engineering today is manual iteration:
  - User cost:
    -Wasted time
    -Reduced trust
    -More retries
  - System cost:
    -Higher token spend
    -Lower retention
    -Increased support load

Prompt → LLM → Output → Score

Mathematical Formulation

$$\min_{\phi} \mathbb{E}_{(x,y)}\left[y \log f_\phi(x) + (1-y)\log(1-f_\phi(x))\right]$$

$$f_\phi(x) = P(\text{human} \mid x)$$

x = generated text output

$y \in \{0,1\}$ = label

- 1 = human-quality / non-slop
- 0 = slop / generic

$f_\phi(x)$ = model's predicted probability that text is human-quality

- Binary classifier trained with cross-entropy:
  - Penalizes confident wrong predictions heavily
  - Encourages calibrated probabilities

Next Steps

$$\max_{\pi_\theta} \mathbb{E}_{p\sim\pi\theta}\left[R(G(p))\right]$$

π_θ = prompt distribution (policy)

p = sampled prompt

G = frozen LLM

R = learned verifier

$G(p)$ = output text

➔ Define a distribution over prompts and optimize it to maximize reward

- ◆ Prompt → LLM → Output → Score

Initial Implementation: Slop vs Human Classification

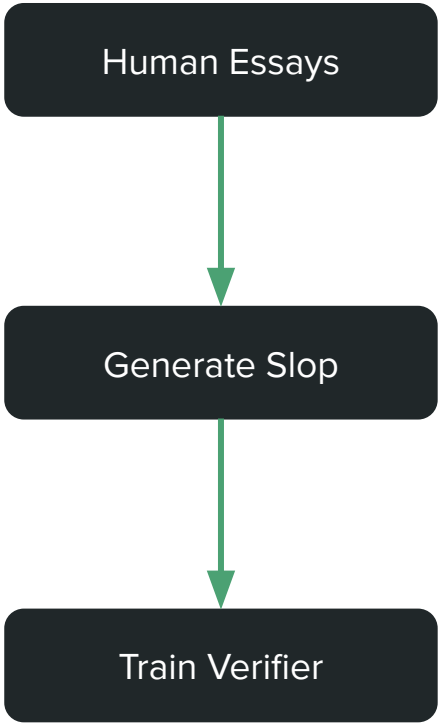
Data:

- Kaggle human essays dataset sourced from Aeon
- Slop generated from title-based prompts using DistilGPT2
- Length filtering applied
- Stratified train/val/test splits

Training:

- Model: DistilBERT binary classifier
- Cross-entropy loss
- Optimizer: AdamW

$$\min_{\phi} \mathbb{E}_{(x,y)} l(f_\phi(x), y)$$



Results

- Test Accuracy: 99.3%
- 2 misclassifications

Most 'human-like' slop (label=0 with highest P(human)):

	p_human	pred	text
126	0.995063	1	I've written some of this because I'm not a fa...
103	0.399538	0	The Mind in the Body\nIf you're reading this o...
203	0.006891	0	A Practical Practical Practical Practical Prac...
154	0.006399	0	Islam. Be careful if you will miss it. If you ...
186	0.004116	0	The Brain Is Made of Nature – The Mind is Made...

Most 'slop-like' human (label=1 with lowest P(human)):

	p_human	pred	text
7	0.093251	0	In 1989, Cambridge University Press announced ...
58	0.998281	1	Parwati Soepangat (1932-2016), fondly known as...
220	0.998422	1	In the past few years I have spent a lot of ti...
87	0.998474	1	The anonymous author of the pamphlet Onania (1...
130	0.998494	1	If the sultan of Mysore had had a bit more luc...

Model achieves near perfect separation...

We need to make the problem harder

What We Learned

- Essays were long (several thousand characters)
- Longer text amplifies repetition and structural artifacts
- Certain models are worse at producing slop
- Classifier may exploit superficial cues

To optimize prompts, we need to understand and improve the slop.

Next Steps: Adversarial Slop and Optimization

1. Train a “sloppifier”: $S_\psi(x) = \mathbf{sloppified}(x)$
2. Adversarial Objective: $\min_\phi \max_\psi \mathbb{E}\left[\log f_\phi(x) + \log(1 - f_\phi(S_\psi(x)))\right]$
3. Optimize Prompts: $J(\theta) = \mathbb{E}_{p \sim \pi\theta}\left[R(G(p))\right]$